Andrew Grathwohl

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SKILLS

LANGUAGES Rust, C, C++, Swift, Node.js, TypeScript, Python, Golang, Elixir.

FRAMEWORKS Next.js, React, Electron, Tauri, Vue 3, Remix, React Native.

AUDIO DSP/ML PyTorch, Demucs, librosa, pedalboard, JACK, FFmpeg, SoX, rubato, rustfft, hound, pyebur128, pyloudnorm, TorchAudio, SpeechBrain, SpaCy.

DB/STORAGE PostgreSQL, Supabase, HAMMER2, GlusterFS, Zfs/OpenZFS, Btrfs, Ceph, IPFS, JuiceFS, XFS.

INFRASTRUCTURE GCP, AWS, Docker, Kubernetes, Terraform, Nix, Ansible, Modal, Vercel.

NETWORKING bird2, FRR, Nginx, Varnish, Netbox, ntop-ng, Kamailio, FreeSWITCH.

SYSTEM TOOLS sysctl, cpupower, bcc/bpftrace, perf, tcpdump, RT_PREEMPT.

OSES GNU/Linux, FreeBSD, DragonFly BSD, macOS, VyOS, NuttX, pfSense, RouterOS, Junos OS.

EXPERIENCE

MAY 2025 - NOVEMBER 2025

SENIOR BACKEND ENGINEER - AUDIO — PARTS ORDER, INC., SAN FRANCISCO, CA (REMOTE)

- Reverse-engineered Dolby Atmos proprietary pan laws and post-processing algorithms through signal analysis, enabling development of ADM-BWF player for both web and TUI interfaces, capable of real-time spatial audio rendering without official SDK licensing - the first implementation of its kind for web and Rust platforms.
- Built production-grade audio mastering limiter in Rust featuring PyTorch gradient optimization for minimizing intermodulation distortion, linear-phase FIR crossover filters, and HRTF-based spatial audio renderer with SOFA format HRTFs and real-time JACK integration.
- Architected production Dolby Atmos mastering platform featuring ML-based stem separation (PyTorch/Demucs/BS-RoFormer), ADM-BWF rendering engine with broadcast compliance, and FastAPI backend with SQLModel/PostgreSQL state management. Implemented this work across Python, Rust, and TypeScript surfaces. codebases.
- Engineered Wine-based integration of Dolby Encoding Engine tooling on Linux infrastructure, enabling Windows-only VST3/AAX plugin processing in production Docker containers with professional loudness measurement and automated gain staging.
- Developed customer-facing web application (Next.js 15/React 19, 201+ TypeScript components) with Supabase auth, Stripe payments, and real-time processing progress tracking.
- Migrated entire infrastructure from bare metal to GCP, optimizing codebase into Docker containers with L4 GPU support.

OCTOBER 2024 - APRIL 2025 — **AUDIO/TELEPHONY CONSULTANT** — HORIZON3 VENTURE STUDIOS, LOS ALTOS, CA (REMOTE)

- Led special projects development for Prudential Insurance call centers, featuring an end-to-end SIP telephony k8s architecture to facilitate AI call center agent deployments across twelve 1-800 numbers.
- Deployed Kamailio and FreeSWITCH SIP services to provide advanced call routing, transfer, conferencing, and SIPREC features in order to satisfy compliance and security requirements.
- Produced custom Linux kernel builds with RT_PREEMPT patches and sysctl tunings optimized for low-latency multimedia processing use cases.
- Developed SIP messaging protocol extension to enable the passing of encrypted customer metadata between call center servers and agent telephones, enabling real-time intelligence and continuous authorization of the caller delivered over WebRTC connections
- SIP telephony and presence implementations integrated the LLM into phone calls directly, leading to a dramatic reduction in call avoidance metrics (between 10-20% drop each day since deployment.)

JULY 2024 - SEPTEMBER 2024 — **SENIOR SYSTEMS ENGINEER - CDN** — RUMBLE, LONGBOAT KEY, FL (REMOTE)

- Architected a KVM-based anycast CDN solution for Rumble's 12 distributed data center locations throughout the United States, responsible for handling an average of 2Tbps traffic during peak usage times
- Generated FRR BGP routing configurations for pfSense and bird2 BGP routing configurations for Ubuntu hypervisor guests, optimizing for the greatest number of active connections per server
- Produced numerous kernel optimizations that enabled Ubuntu guests to serve cached assets with a 18ms reduction in round-trip travel times
- Implemented a custom Rust-based caching service to replace nginx and openresty caching services, enabling Rumble's live streaming servers to exclusively leverage in-house hardware to route RTMP, WHIP, and SRT broadcasts to end users
- Developed inventory analysis scripts to assess current server memory footprints, which automatically produced purchase orders for additional memory DIMMs based on a provided capacity target

DECEMBER 2023 - PRESENT — **FOUNDER/CEO** — TUBE.SH, LLC, NASHVILLE, TN (REMOTE)

- Launched LLC to centralize contract work in the media services, video analytics, and content operations domains. The business offers "Patreon in a box" services to independent (and sometimes controversial) media businesses and their creative talent. Leading a small team of contractors for design, security, and business management consultation while I build main open source code bases and handle infrastructure deployment & instrumentation.
- First project is development and operation of *PartOfTheProblem.com*, the premium subscription service for Dave Smith's popular long-running political podcast (150k downloads per episode, $35,000 MRR).
- Notable backend work items include: archiving and ingestion of 3TB of video content and all metadata, custom RTMP ingest endpoints hosted within a Docker Swarm and fail-over support across three separate CDN providers (CDN77, CloudFlare, AWS CloudFront), multiple tiers of usage including a free tier and 5 paid tiers, mezzanine video upload portal with FFmpeg transcoding and SoX audio conversion, private podcast feed implementation with strong RSS 2.0 compliance and password protection for each paying subscriber, multi-CDN weighted round

robin video streaming implementation, full monitoring and alerts dashboard via Grafana, Prometheus, and Sentry.
- Notable full-stack work items include: self-hosted PayloadCMS service for user-friendly website editing, podcast feed modifications, and content publishing. Nuxt3 progressive web application, hosted on Vercel with critical K/V stores and serverless functions hosted on the edge.

AUGUST 2020 - SEPTEMBER 2023

LEAD ENGINEER — STORYBOARD, NEW YORK, NY (REMOTE)

- Bootstrapped and mentored Storyboard's audio-first engineering team as the company's first engineering hire, growing headcount from 1 to 16 in one year.
- Oversaw and implemented CI/CD workflows, code quality standards, and cloud architecture for the world's largest private podcasting network for enterprises and its companion audio chat service for front-line and first responder workforces.
- Developed Storyboard's machine learning pipelines, responsible for producing transcripts from spoken word recordings at 98% accuracy. Devised AWS SageMaker pipeline to parse entities in transcripts and separate concepts. Utilized temporal convolution and chromaprint to detect authenticity of recorded voices.
- Invented a novel approach to achieving sub-30ms latency audio communication channels over a swarm WebRTC decentralized network, with STUN/TURN signaling at both the client and within a cluster of centralized publicly-available WebTorrent nodes.
- Implemented company's iOS, tvOS, and CarPlay audio frameworks as Swift Packages, providing support for Opus transcoding, real-time audio classification with CoreML, and live Whisper transcript features.
- Implemented AWS Lambda low-latency audio analysis and transcoding pipeline for validating uploaded content and generating RF64 masters and opus encodes.
- Developed recording, editing, and playback suite consisting of Web Audio API modules and Rust-WASM binaries, enabling generalist engineers to build audio creation tools that perform complex audio mixing and DSP operations.
- Led development of white label app for Amazon.com, enabling the distribution of private podcast episodes to Amazon's 10,000+ fleet of delivery vehicles.

AUGUST 2017 - JUNE 2020

DIRECTOR - MEDIA TECHNOLOGY — LITTLSTAR, NEW YORK, NY

- Launched, managed, and mentored the Media Services remote engineering team in pursuit of executing media technology strategy. Carried out hiring, budget, sprint planning, code review, and performance tracking responsibilities.
- Technical business development lead for enterprise contracts signed with AT&T 5G, Intel, Microsoft, ViacomCBS, Sony, MyndVR, WWE, and Live Nation.
- Developed and implemented multi-CDN Node.js services responsible for ingesting, processing, tracking, and validating 360/3D video, VR experiences, traditional SVOD media, RTMP live streams, and volumetric holograms. Performed frequent benchmarks and performance analyses to ensure reliable low-latency peer-to-peer connectivity for all services.
- Produced novel implementations of new MPEG standards with the goal of improving streaming QoE, including CUDA transcode acceleration, tile-adaptive MPEG-DASH, real-time object detection neural nets, and peer-to-peer distributed encoding.
- Co-creator of Ara, a decentralized blockchain-backed content distribution proof-of-stake cryptocurrency which rewards peer-to-peer sharing of media content. Architect of the Ara File Manager electron application.

APRIL 2016 - AUGUST 2017

LEAD MEDIA ENGINEER — LITTLSTAR, NEW YORK, NY

- Led the integration of proprietary Littlstar video player technologies into business partners' products, including Alcatel, CNN, and Mattel.
- Liaised with content partners to promote success on the platform, providing documentation and hands-on technical support to ensure compliant media assets and strong content performance KPIs.
- Developed Littlstar's media library standards, including audiovisual QC workflows, livestream setup requirements, digital rights management, metadata internationalization, storage and permissions management, and content taxonomy.

JANUARY 2016 - APRIL 2016

SENIOR BUSINESS ANALYST - VIDEO — CBS CORPORATION - SHOWTIME NETWORKS INC., NEW YORK, NY

- Evaluated business requirements, technical processes, and operations improvements necessary to bring Showtime Networks' digital asset management services in-house.
- Identified and tracked key performance indicators for content management operations, presenting weekly reports to CBS IT executive leadership.
- Spearheaded initiative to conduct regular transfer speed tests for hard drive, solid-state, and NAND storage media, increasing the accuracy of CBS IT's predictive models for storage requirements.
- Modernized Showtime Networks' content metadata taxonomy, optimizing for greater interoperability throughout the CBS organization.

SEPTEMBER 2015 - DECEMBER 2015

MEDIA ENGINEER — STREAMABLE [CONTRACT], BROOKLYN, NY

- Assessed the company's video transcoding pipeline and identified key improvement opportunities to better target their growing mobile user base.
- Addressed bottlenecks in video processing services that prevented timely delivery of high-frame rate content (particularly sports and gaming content) to mobile devices.
- Updated chrome browser plugin that Streamable's users leveraged to make real-time clips of their video gaming sessions to share among their social media connections.

APRIL 2015 - SEPTEMBER 2015

BACKEND ENGINEER — OVERTURE.ME, NEW YORK, NY

- Developed the video editing backend for Overture.me, a video sharing social media app for iOS. The backend acquired video editing instructions from the mobile application and performed editing, concatenation, and final authoring of video content for rapid distribution to users' networks of followers.
- Implemented custom FFmpeg concatenation filters that enabled videos to be published and edited within 30 seconds of their initial submission.

DECEMBER 2014 - APRIL 2015

AUDIO SCIENTIST — SPOKENLAYER [CONTRACT], NEW YORK, NY

- Developed the SpokenLayer "cloud studio," consisting of a web app for recording and editing spoken word audio, a real-time bidding platform for new narration jobs, and a backend processing pipeline to convert and process audio recordings for distribution.
- Integrated microservices into a standardized workflow which enabled audio narrations for news articles to be published to client websites within 30 minutes of the text's publication.
- Head of production for the Stitcher Top 5 Podcast, TIME's The Brief, and Reuters World News Report, which yielded a combined 500k listeners per day.

SEPTEMBER 2013 - DECEMBER 2014

ACX PRODUCTION COORDINATOR — AUDIBLE, INC., NEWARK, NJ

- Managed Audible's Audio QA team, consisting of four full-time audio engineers, responsible for validating all Audible titles prior to publication.

- Director of ACX audio operations, including ingestion, production guidelines, QA workflows, and transcoding.
- Scaled ACX's production volume from 200 audiobooks per month to over 3,000 audiobooks per month.
- Introduced backend audio quality metrics and QA validation results into Amazon's internal dashboard framework.
- Inventor of Audible voice search technology, US Patent 9412395: *Narrator selection by comparison to preferred recording features*.
- Regularly blogged about and presented audio production techniques under the name "Andrew the Audio Scientist."

OCTOBER 2011 - SEPTEMBER 2013

POST-PRODUCTION ASSOCIATE — AUDIBLE, INC., NEWARK, NJ

- Edited and produced more than 150 premium Audible Studios audiobooks.
- Defined department-wide audio QA guidelines and encoding specifications.
- Developed an AWS web service which re-formats Kindle book files into screenplay-style narration scripts for voice actors.

EDUCATION

AUGUST 2007 - JULY 2011

BACHELOR'S OF SCIENCE, RECORDING ARTS — INDIANA UNIVERSITY, JACOBS SCHOOL OF MUSIC, BLOOMINGTON, IN

Computer Music Concentration